

Problems marked with a (*) are a bit harder. Hints are available in the final page.

1.1. Recap Questions.

1. Let $\{x_k\}_{k \in \mathbb{N}} \subset [1, 2] \subset \mathbb{R}$ be a sequence. Is it necessarily convergent?
2. Describe all differentiable functions $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ such that $f'(x) = 1/x$ for all $x \neq 0$.
3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous and bounded, is it true that

$$\int_0^{\pi/2} f(\cos(x)) dx = \int_0^1 \frac{f(t)}{\sqrt{1-t^2}} dt?$$

4. Let $f: [0, \infty) \rightarrow \mathbb{R}$ have $f''(x) > 0$ for all $x \geq 0$. Can f have a global maximum point?
5. Let $\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a linear map and let $Q := [0, 1] \times [0, 1]$ be the unit square. How does $\phi(Q)$ look like? What are the possibilities?
6. Consider the vector space V of polynomials of degree less than 4, and the function $D: V \rightarrow V$ which associates to a polynomial p its derivative p' . Is D a linear map? If so, what is the dimension of its null space?

1.2. From equations to drawings. Sketch the following subsets of $\mathbb{R}^2, \mathbb{R}^3$.

1. $A := \{(x, y) : x^2 + y^2 < 1, y \geq x^2\}$
2. $B := \{(x, y) : y^2 = x^3 - x\}$
3. $C := \{(x, y) : \frac{\pi}{6} < \arctan(y/x) < \frac{\pi}{3}, 0 < x < y\}$
4. $D := \{(x, y, z) : \sqrt{x^2 + y^2} = 1 - z, 0 < z < 1\}$
5. $E := \{(x, y, z) : z = \sqrt{1 + x^2 + y^2}\}$
6. $F := \{(x, y, z) : z = \max\{|x|, |y|\}\}$

1.3. From drawings to equations. Express in terms of functions and equations the following sets.

1. The coordinates of a point in the plane that goes endlessly back and forth from the origin to the point $(-1, 1)$.
2. A planar domain with two holes in it.
3. A bi-dimensional disk sitting inside the plane $\{x + y + z = 0\}$ in $\mathbb{R}^3$.
4. A Pac-Man shaped domain in the plane.
5. A pyramid, whose base is a square, sitting in $\mathbb{R}^3$.

1.4. Young's inequality. Let $a, b > 0$, and let $p, q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$. The goal of this exercise is to prove that $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$.

Obs: The case $p = q = 2$ corresponds to the well-known inequality $ab \leq \frac{a}{2} + \frac{b}{2}$, which is just a rewriting of $(a - b)^2 \geq 0$.

1. Prove that for all $t > 0$,

$$t^{1/p} \leq \frac{t}{p} + \frac{1}{q}.$$

2. Multiply the inequality by b^q and choose an appropriate value of t to conclude.

A bonus: Show that $\langle a, b \rangle \leq \frac{\|a\|^p}{p} + \frac{\|b\|^q}{q}$ for all $a, b \in \mathbb{R}^n$.

Hints:

- 1.2 As ETH students you have a license for Mathematica. All these plots are one-line-commands using that software!
- 1.2.2 The set is clearly symmetric with respect to $y \mapsto -y$, so it's enough to plot qualitatively the function $y(x) = \sqrt{x^3 - x}$, separately in the intervals $x \in [-1, 0]$ and $x \in [0, \infty)$... From here you can use all the technology from Analysis I...
- 1.2.4 Try to figure out how D looks layerwise, i.e., as a planar domain once z is kept fixed.
- 1.2.5 If you know how to plot $z = f(s), s \geq 0$, then the set $\{(x, y, z) : z = f(\sqrt{x^2 + y^2})\}$ is obtained rotating the plot of f around the z -axis...
- 1.3.1 Construct an appropriate function $f: \mathbb{R} \rightarrow \mathbb{R}^2$, where t represents the time and $f(t)$ represents the position of the moving point at time t . You want f to be periodic in t and $f(t)$ to be aligned with the direction $(-1, 0)$
- 1.3.3 Observation: the intersection of a round ball with a plane is a disk...
- 1.3.5 Take inspiration from 1.2.4 and 1.2.6...
- 1.4.1 Defining $g(t) = \frac{t}{p} + \frac{1}{q} - t^{1/p}$, this is equivalent to $g(t) \geq 0$. Compute $g'(t)$.

1. Solutions

Solution of 1.1:

- 1. No, take for example $\{3/2 + (-1/2)^k\}_{k \in \mathbb{N}}$. What is true is that by the Heine-Borel Theorem, $\{x_k\}_{k \in \mathbb{N}}$ has a convergent *subsequence*.
- 2. f is determined up to a constant in each of the intervals $(-\infty, 0)$ and $(0, \infty)$. The point is that these two constants may be different, hence the most general f is

$$f(x) = \begin{cases} \alpha + \log x & \text{if } x > 0, \\ \beta + \log(-x) & \text{if } x < 0, \end{cases}$$

for some $\alpha, \beta \in \mathbb{R}$. Remark: the answer $f(x) = \log |x| + c, c \in \mathbb{R}$ is *wrong*.

- 3. Yes, this is the change of variables formula in the integrals.
- 4. Yes, $f := e^{-x}$ has a maximum point at $x = 0$ and $f'' = f > 0$.
- 5. $\phi(Q)$ is the parallelogram generated by the two vectors $\{\phi(1, 0), \phi(0, 1)\}$. To be precise, if $\phi(1, 0), \phi(0, 1)$ are aligned the parallelogram degenerates in fact into a segment, and if $\phi(1, 0) = \phi(0, 1) = 0$ then it degenerate (even more) into the single point $\{0\}$.

6. Yes, the derivative is a linear operator when acting on functions; in particular, it stays a linear operator when it acts on polynomials. V has dimension 4 since it is generated by the basis $\{1, X, X^2, X^3\}$ (this is, ultimately, the very definition of a polynomial of degree less than 4). One explicitly computes that, via D ,

$$1 \mapsto 0, \quad X \mapsto 1, \quad X^2 \mapsto 2X, \quad X^3 \mapsto 3X^2.$$

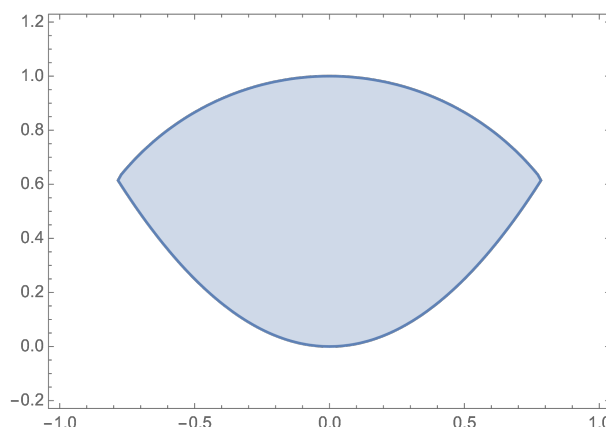
In matrix form (with respect to the basis $\mathcal{B} := \{1, X, X^2, X^3\}$) this reads:

$$[D]_{\mathcal{B}, \mathcal{B}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The rank (=dimension of the span of the columns) of this matrix is 3, so by the rank-nullity Theorem, the nullity is $4 - 3 = 1$.

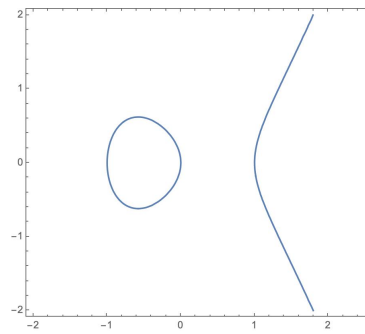
Solution of 1.2:

1. One has to intersect the disk $\{x^2 + y^2 \leq 1\}$ with the epigraph of the parabola $\{y = x^2\}$:

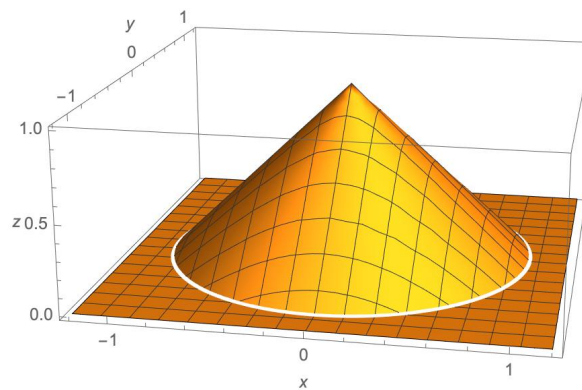


$$\{(x, y) : x^2 + y^2 < 1, y \geq x^2\} \cap [-1, 1] \times [-0.2, 1.2]$$

2. Since $x^3 - x = x(x+1)(x-1) \geq 0$ if and only if $x \in [-1, 0] \cup [1, \infty)$, one needs to study the function $f(x) := \sqrt{x^3 - x}$ separately in these two intervals. The set B will then be the union of the graphs of f and $-f$. In the interval $[-1, 0]$, f is concave and vanishes at the extrema, with a maximum point at $x = -\frac{1}{\sqrt{3}}$. Also in $[1, \infty)$ f is concave and monotonically increasing, vanishing at $x = 1$. At infinity it grows like $x^{3/2}$.
3. Since $0 < x < y$, the set C is contained in the second octant of the plane. The condition $\frac{\pi}{6} < \arctan(y/x) < \frac{\pi}{3}$ forces any point in C to have span an angle between 30° and 60° with respect to the positive x axis. In summary we find that C is the region of the first quadrant of the plane between the lines $\{y = x\}$ and $\{y = \sqrt{3}x\}$.

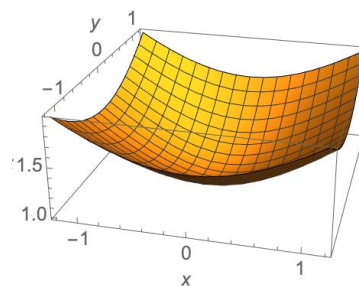
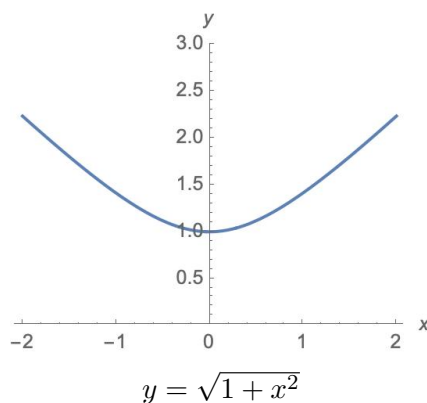


$$\{y^2 = x^3 - x\} \cap [-2, 2.5] \times [-2, 2]$$



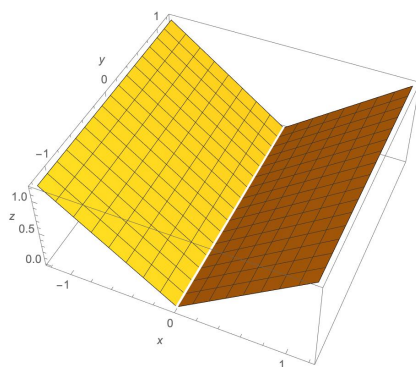
$$\{\sqrt{x^2 + y^2} = 1 - z\} \cap [-1.2, 1.2] \times [-1.2, 1.2] \times [0, 1]$$

4. The set $D \cap \{z = t\}$ is a circle of radius $1 - t$, so the picture must be a round cone of height 1 over the unit circle:
5. As in the Hint, the set is obtained rotating the plot of $y = \sqrt{1 + x^2}$ around the z axis:

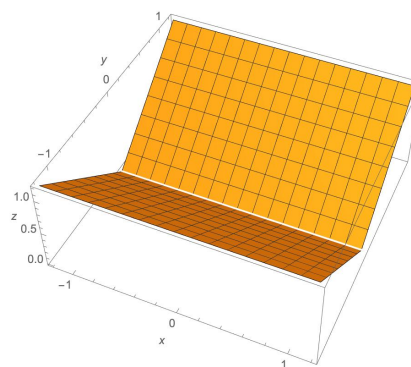


$$\{z = \sqrt{1 + x^2 + y^2}\}$$

6. First one plots in 3D $z = |x|$ and $z = |y|$, obtaining:

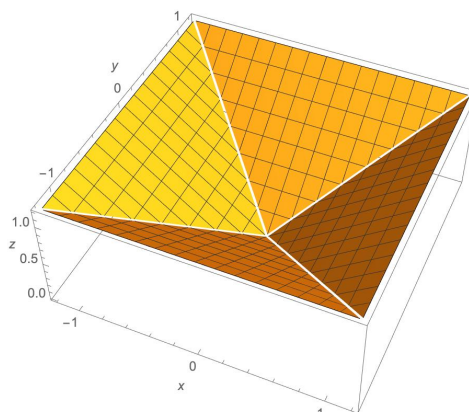


$$\{z = |x|\}$$



$$\{z = |y|\}$$

Then one takes the maximum of the two plots. To visualize this, it might be helpful to notice that the level sets $\{\max\{|x|, |y|\} = t\}$ coincide with the squares $[-t, t] \times [-t, t]$, for all $t \geq 0$. One finds



$$\{z = \max\{|x|, |y|\}\}$$

Solution of 1.3:

1. A point that moves along the line spanned by $(-1, 1)$ is represented by $\mathbb{R} \ni t \mapsto (-f(t), f(t)) \in \mathbb{R}^2$, where $f: \mathbb{R} \rightarrow \mathbb{R}$ is some function. We want the motion to take place in the segment, so we require $0 \leq f(t) \leq 1$. We also want the function to be defined for all $t \geq 0$ (“endlessly”). We also want the function to go back and forth, so a natural guess at this point is $f(t) = \sin(t)$... but of course other choices are possible.
2. Remove two small squares from a large square

$$\{10 > \max\{|x|, |y|\} > 1, \max\{|x - 5|, |y - 6|\} > 1\}.$$

3. As suggested it is enough to intersect the ball $\{x^2 + y^2 + z^2 \leq 1\}$ with the plane $\{x + y + z = 0\}$.

4. Remove from a disk a 30° angle:

$$\{x^2 + y^2 \leq 1\} \setminus \{|\arctan(y/x)| \leq \pi/6, x > 0\}.$$

5. We saw that the level sets of $(x, y) \mapsto \max\{|x|, |y|\}$ are squares (cf. 1.2.6), and we saw how to construct a pyramid using a function whose level sets were circles (cf. 1.2.4). Combining the two formulas one gets

$$\{1 - z = \max\{|x|, |y|\}, 0 \leq z \leq 1\}.$$

Solution of 1.4:

1. Note that $g(1) = 0$ and

$$g'(t) = \frac{1}{p} \left(1 - t^{-(p-1)/p}\right).$$

Considering the sign of $g'(t)$, $t = 1$ is the global minimum of $g(t)$ and hence $g(t) \geq 0$.

2. Multiplying by b^q we obtain

$$t^{1/p} b^q \leq \frac{t b^q}{p} + \frac{b^q}{q}.$$

Considering $t = \frac{a^p}{b^q}$ we find

$$a b^{q-q/p} \leq \frac{a^p}{p} + \frac{b^q}{q},$$

which since $q - q/p = q(1 - 1/p) = q \cdot \frac{1}{q} = 1$ concludes the proof.